



JAIDEV EDUCATION SOCIETY'S
**J D COLLEGE OF ENGINEERING AND
MANAGEMENT**
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



B. Tech Winter - 2026 (ESE) Examination

**Branch: Computer Science & Engineering
Year)**
Subject: Elective -V (React JS)
Marks: 100

**Year/Semester: 5th Sem (3rd
Subject Code: CS5E002C**

Question Bank

State the meaning of Route Params in React Router v6.

.Identify the purpose served by the useContext () hook in React applications.

List two applications of LocalStorage in React development.

Differentiate between **data passing through Props** and **data sharing through Context API**.

Illustrate the concept of **Nested Routes** in React Router v6 with an example.

Discuss the process of maintaining **loading states** while integrating APIs in React.

Analyze the methods of component communication through **Props, Context API, and Custom Hooks**, and conclude the most suitable approach for large-scale applications.

Evaluate the design of a **product listing feature** in a React application with reference to **Search, Filters, and Pagination**, and justify the approach that enhances user experience.

Create a functional React component to integrate API data using **Axios**, implement **loading and error handling states**, and display the output in a structured list format.

Define React and explain its role in building user interfaces.

State the purpose of using the Virtual DOM in React applications.

Identify the difference between class components and functional components in React.

List out the key advantages of using JSX in a React project.

Rewrite the steps for updating a component's state using the `setState()` function in class components.

Classify the different types of component states used in React.

Restate the concept of event handling in React with an example.

Write the syntax of the `useEffect()` hook and describe its purpose.

Give an example of using the Context API for data sharing between components.

Define the concept of asynchronous programming and its role in API integration.

State the main purpose of using environment variables in a React project.

List the essential steps involved in deploying a React project on Netlify or Vercel.

Distinguish between controlled and uncontrolled components in React with suitable examples.

Illustrate the working of the `useEffect()` hook with an example showing data fetching.

Discuss how event handling differs in class components and functional components.

Examine the concept of props drilling and how Context API helps to overcome it.

Write a short note on nested routing using React Router v6.

Sketch a simple diagram showing data flow through props and context in a React application.

Illustrate the working of the `useContext()` hook with a suitable example that demonstrates global state management.

Explain the flow of data between parent and child components using props in a routing-based application.

Analyse how API error handling and loading states contribute to improving user experience in React apps.

Examine the concept of LocalStorage and discuss its importance in maintaining data persistence.

Describe the key steps in building and structuring a complete React project (e.g., Blog or E-Commerce UI).

Discuss how code organization, modular components, and deployment practices impact long-term project maintainability.

Rewrite about following:

- a) . Passing data via props
- b) . Usage of `useContext()` hook
- c) . Component communication hierarchy
- d) Role of Custom Hooks in reusability

Explain asynchronous programming concepts and demonstrate how Promises and `async/await` are used in API calls.

Describe different performance optimization techniques like code splitting, lazy loading, and memoization in React.

Define JSX and explain how it helps in writing React components.

State the difference between Real DOM and Virtual DOM.

Identify the importance of props in component communication.

List out the main steps to create a new React application using Create React App.

Rewrite the syntax of the `useState()` hook and explain how it simplifies state management.

Classify the various phases of the React component lifecycle.

Restate how event handling is implemented in functional components.

Write the syntax for declaring and updating a state variable using `useState()`.

Give an example of using `useContext()` to manage global state.

Define loading states in React and explain their importance during API calls.

State the role of code splitting in improving React application performance.

Enlist the best practices for optimizing React app performance before deployment.

Distinguish between stateful and stateless components in React with examples.

Illustrate how form handling is implemented using the `useState()` hook.

Discuss the importance of lifecycle management using hooks in functional components.

Examine the usage of `useContext()` for managing global data within nested components.

Write a short note on the role of route parameters in React Router v6.

Sketch the structure of a simple React application showing parent and child component communication.

Illustrate the working of navigation and route parameters in React Router v6 with an example.

Explain how Custom Hooks can be designed to simplify state management and data fetching in React.

Analyse the process of integrating APIs using Axios and compare it with the Fetch method.

Examine the role of filters and search functionality in improving data accessibility for users.

Describe best practices for styling React components using CSS Modules and Styled Components.

Discuss the deployment workflow of React projects on cloud platforms and explain the importance of environment configuration.

Rewrite about following:

- a) React Router v6 architecture
 - b) Route parameters and nested routes
 - c) Navigation using `Link` and `NavLink`
 - d) Role of Context API in data sharing between components
- Explain** the process of API integration in React applications using `fetch()` and `axios` with suitable examples.
- Describe** the complete workflow of developing and deploying a React project on Netlify or Vercel

Define React and explain its role in building user interfaces.

State the purpose of using the Virtual DOM in React applications.

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Discuss how code organization, modular components, and deployment practices impact long-term project maintainability.

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a) . Passing data via props

b) . Usage of `useContext()` hook

c) . Component communication hierarchy

d) Role of Custom Hooks in reusability

Explain asynchronous programming concepts and demonstrate how Promises and `async/await` are used in API calls.

Describe different performance optimization techniques like code splitting, lazy loading, and memoization in React.

Define a React component and mention its basic structure.

State the significance of keys when rendering lists in React.

Identify the role of React Developer Tools in debugging applications.

List out different ways to style components in React.

Rewrite the process of handling form submission events in React functional components.

Classify the difference between controlled and uncontrolled components.

Restate the working of `useEffect()` hook with cleanup functions.

Write a short note on conditional rendering in React.

Give an example to show how data is passed through props between components.

Define pagination in the context of displaying API data on a React web page.

State the steps required to deploy a React application on Netlify.

Enlist various deployment platforms available for React projects.

Distinguish between `setState()` in class components and `useState()` in functional components.

Illustrate the concept of conditional rendering with a suitable code example.

Discuss the use of the `useEffect()` hook to manage side effects in React.

Examine the working of React Router and its contribution to single-page applications.

Write a short note on the Context API as a tool for component communication.

Sketch a component hierarchy showing how data flows from parent to multiple child components using props.

Illustrate with an example how React Router v6 is used to implement nested and dynamic routing in a React application.

Explain the process of sharing data between components using Context API and `useContext()` hook.

Analyse the importance of asynchronous programming in API integration and compare the use of Promises and `async/await`.

Examine how pagination, search, and filtering enhance the usability and performance of a React application.

Describe the steps involved in deploying a React application on Netlify and managing environment variables.

Discuss different methods of performance optimization and code splitting in large-scale React projects.

Rewrite about following:

a) Route configuration in React Router v6

b) Navigation between pages

c) Context API workflow

d) Custom Hook creation and usage

Explain how to handle API errors, manage loading states, and implement pagination in real-world React projects.

Describe the process of styling React applications using CSS Modules and Styled Components, and explain their advantages.

Define React components and state their role in UI composition.

State two advantages of using Virtual DOM over Real DOM.

Identify the main difference between props and state in React.

List out four core features offered by React library.

Rewrite the steps to create a functional component that uses the use State hook.

Classify different types of events supported in React (e.g., keyboard, mouse, form).

Restate how the use Effect hook helps simulate component lifecycle behaviour.

Write short notes on the concept of conditional rendering in React.

Give an example of using Context API for sharing data between multiple components.

Define pagination and state its purpose in data-driven web applications.

State how environment variables help secure API keys in React projects.

List any three optimization techniques used to improve React application performance.

Distinguish between use State and use Reducer hooks in terms of state management.

Illustrate form validation in React using controlled components.

Discuss how lifecycle methods in class components are replicated using React hooks.

Examine the impact of conditional rendering on user interface performance.

Write short note on route parameters and their usage in React Router v6.

Sketch a component tree diagram showing how props flow through nested components.

Illustrate the working of nested and dynamic routes using React Router v6.

Explain how custom hooks enhance code reusability and simplify API data fetching.

Analyse the importance of search and filter features in React data-driven applications.

Examine how Local Storage and Session Storage are used to preserve data state.

Describe how to organize React components and apply styling using CSS Modules and Styled Components.

Discuss how code splitting and lazy loading improve large-scale React project performance.

Rewrite about following:

1) React Router v6 architecture

2) Custom Hook structure

3) Nested Routing workflow

4) Context API usage

Explain the complete process of API integration using Axios /fetch with error and loading handling.

Describe the end-to-end workflow of building, optimizing, and deploying a React project on Netlify or Vercel.